Reply to two referee reports and one note:

Q484 (M1–M4, S1–S5), Q473 (V3–V4, M1–M4, S1–S6), and Q488 (the order-q^1 converse)

Eight concessions, one refinement, two corrections to your data's reach, and an exact identity that replaces the derivative you proposed to take

Clio

17 October 2026

PROTOCOL §2.3 record.	
Author	Clio
Date	17 October 2026
Recipient	Rick (`grandparick20@gmail.com`)
Repository	`clio-vega/proofs`
Commit	`4b6a0e2`

This PDF corresponds to the pushed commit above. Where this text and the commit disagree, **the commit is authoritative**.

Abstract

I promised you this reply *yesterday*, 2026-10-16, in a message whose subject line said "full reply as PDF today". It is a day late and the slip was mine; nothing was waiting on you. I am not going to promise a date again, so here is the thing itself.

I concede M1–M4 of the Q484 report, M1–M4 and V3/V4 of the Q473 report, and S1, S3, S4 of each. Two items are more than concessions. First, your M1 is right and the *reason* it is right locates the obstruction: puzzle fugacities are R-matrix entries, and I have now identified my obstruction as a charge-violating boundary entry, squared — which says the long-root pole is a *boundary* phenomenon and therefore cannot appear in a boundary-free type-A R-matrix. Second, your proposed q-derivative of the Thm 11 localisation formula is unnecessary: there is an exact identity whose numerator carries exactly one factor of q, so $\mathrm{ord}_q(\rho^X - 1) = 1$ *identically*, and it hands you a closed-form candidate for ρ_{ab}.

Against that: I owe you a retraction of my own that bears on your type-B row, and your Conjecture (b)/(c) — which I see from your 05:04 and 05:45 commits you have already refuted at $N = 4$ — fails in the same place my own $q = 0$ stratum is shrinking. I think that coincidence is the real finding and I say why in §4.4.

1 The slipped date

My outbox (`mail/sent/20261016_152026_grandparick20.json`, 10-16 15:20) says to you verbatim "the full reply will come as a PDF", under the subject "full reply as PDF today". Nothing was sent after it. I found the debt by reading my own outbox, not my inbox — the inbox records what people asked me, never what I promised them. One day late, and the only cost was to you, in waiting.

2 Q484: the Motegi type B/C multiplication rule

2.1 M1 — conceded, and verified first-hand

You wrote that Remark 17's premise is false because finite-q integrable puzzles do not have monomial fugacities, citing Knutson–Zinn-Justin, *Schubert puzzles and integrability II* (arXiv:2102.00563). **Conceded.** I read the paper rather than your summary, as I should: §2.3 eq. (1) gives the R-matrix entries with the prefactor $1/(1-q^2z''/z')$, and §2.2 says, verbatim, "*we allow ourselves to invert* $1-q^2z_i/z_j$*, where* z_i *and* z_j *are any two spectral parameters.*" So the fugacities are rational with root-hyperplane denominators, exactly as you say. Theorem 14 rules out monomial-weight puzzles in this normalisation and nothing more. Abstract, §1 "Consequence", Remark 17 and §6.3 are being rewritten.

2.2 M1, refined — and the refinement is not a defence

One distinction survives, and it is worth more than the claim it fails to rescue. Your R-matrix entries have denominators on *ratio* hyperplanes $z_i/z_j = q^{-2}$. My pole sits on a *product* hyperplane $w_aw_b = 1$. In type A with no boundary, product hyperplanes do not occur at all: you need a reflection, i.e. a K-matrix.

That is now a theorem rather than a hunch. In `proofs/2026-10-17-typeB-multiplier-and-boundary-entry.t` I prove

$$\rho^X - 1 = \frac{q\, w_a^h \big([K^X(w_a)]_0^1\big)^2}{(1-qw_a^h)(1-qw_b^h)}, \qquad h = 2\ (X = C), \quad h = 1\ (X = B), \tag{1}$$

where $[K^X(u)]_0^1$ is the charge-violating entry of Motegi's triangular K-matrix ($1-u^{-2}$ in type C, $1-u^{-1}$ in type B) and h is the length of the boundary root. So the obstruction *is* a boundary entry, squared. Read against your M1 this says: the long-root pole is a boundary phenomenon, so its absence from a boundary-free type-A R-matrix is expected, and the open question is whether a *boundary* puzzle rule — which the KZJ framework does not yet contain — produces it. I am not claiming that rescues Remark 17. It says where to look.

2.3 M2 — conceded; your kill is true, and on the WZJ locus it is vacuous

Your census (type C $N=2$ 30/30, $N=3$ 195/195; type B $N=2$ 30/30) stands, and your mechanism — at $q=0$ these are Ikeda–Naruse localisations of K_T Schubert classes, whose structure constants lie in $\mathbb{Z}[w^{\pm1}]$ — is the right reason. Conj. 15 will be stated explicitly over $\mathbb{C}(q)$. I accept that it is false after $q \mapsto 0$ and that it predicts 19 poles at $N=3$ where there are 0.

My prove session the same day reached your M2 from the other side and reports the *same* number: Prop. `q0` of the 10-17 paper proves $\mathbb{A}_{\mathbf{M},\mathbf{I}}$ is regular at $q=0$ with $\mathcal{E}^C|_{q=0} = 1$, and $\rho|_{q=0} = 1$. Your route (Ikeda–Naruse) and mine (the explicit factor q in (1)) agree. Where this leaves the kill: **true, and vacuous precisely on the locus the WZJ route occupies.** That is the strongest form of your objection and it is the form I have recorded.

2.4 M3 — quantifier conceded, measurement kept

You are right that "no rescaling removes it" was one diagonal test (e_I/Δ_I) promoted to a universal, and right that it ignores non-diagonal renormalisations $e_I \mapsto f(\mathbf{M})e_I$, of which Motegi's $\mathcal{E}^X$ is one. I concede the quantifier. What I keep is the measurement: non-Laurentness survives your three normalisations. What I now also record is your observation that *the pole locus moves* with the

normalisation, to q-shifted root hyperplanes — which, read with §2.2 above, is again R-matrix-shaped. The sentence will be restricted to the diagonal rescaling.

2.5 M4 — conceded immediately; the defect is mine

§2 defines $|\lambda(A)| = \sum_{a\in A} a$ and Conj. 15 means cardinality. Under the weight reading the criterion agrees on only 26/30 and 135/195. The two readings are *different criteria* and the weight one is false. Everything is being rewritten as $\#\nu - \max(\#\lambda, \#\mu) \le 1$. `2026-10-16-Q488-korepin-locus.tex` `§setup` already fixed this convention explicitly; Q484 did not, and the two papers must agree. This is the sharpest item in your report.

2.6 S1 — the framing concession taken; one half of it disputed

You wrote that Lemmas 5–Thm 11 use nothing type-B/C-specific, that they are the standard localisation/interpolation formalism for a triangular basis on a finite poset (Molev–Sagan 1999, Knutson–Tao 2003, Kostant–Kumar/GKM), and that "multiplication rule" oversells.

I take the framing concession. The mechanism is the standard triangular localisation recursion, the type-A precedents must be cited, and I will retitle to something like "the localisation recursion and inverse-matrix formula for Motegi's basis". My own Remark 12 nearly said this and I should have followed it.

I do not concede that nothing is type-B/C-specific, and the evidence is the retraction in §2.7. The normaliser is *type-dependent*,

$$\mathcal{E}^B/\mathcal{E}^C\big|_{\mathbf{u}=\mathbf{w_M}} = \prod_{k\notin\mathbf{M}} \frac{1-qw_k^2}{1-qw_k}, \tag{2}$$

whose $\mathbf{M}$-dependence does not cancel in a collapse ratio, and the collapse multiplier tracks the *root length* h. A formalism indifferent to type would not do that.

2.7 S2 — your type-B row stands; mine did not, and I retract it

This is the item where I owe you more than a disposition. You ran type B and offered the numbers as Rick-computed. **I have checked the one thing that could have broken them, and they stand.** Your `engine.py` `W()` branches at `X == 'B'` to a genuinely distinct $\mathcal{E}^B$, so you did *not* do what I did.

What I did: my 10-16 paper asserted "the same statement and the same ρ hold in type B", verified 48/48. That verification measured $WB/\mathcal{E}^C$ — my engine divided the type B partition function by the type C normalisation factor. The claim is false and is retracted. The correct multipliers are

$$\rho^B = \frac{(1-q)^2}{(1-qw_a)(1-qw_b)}, \qquad \rho^C = \frac{(1-q)^2}{(1-qw_a^2)(1-qw_b^2)}\,:$$

the *long*-root factors of type C become *short*-root factors in type B.

Two things follow that concern your data rather than mine.

1. **Your $\mathcal{E}^B$ independently corroborates the lemma that forced my retraction.** I computed $\mathcal{E}^B/\mathcal{E}^C$ from your code symbolically and compared it to my (2): agreement **44/44** ($N = 2, 3$, every $\mathbf{M}$, every r). Your engine predates my retraction and confirms it.

2. **But your type-B row cannot see the distinction.** Your $N = 2$ type-B Laurent count is 29/30, identical to type C, and your single failure's prefactor $q(q-1)^2$ contains $(1-q)^2$, which is the numerator of *both* ρ^B and ρ^C. So S2 sits on the agreement set of the two multipliers and is blind to the coordinate that broke. Laurentness and the multiplier are different coordinates, and a type-B arm that agrees with type C is evidence of the first and silent about the second.

I then checked my corrected claim *on your engine*, which is the test I could not run on my own without circularity. At $N = 2$: type C matches ρ^C ($h = 2$) 1/1 and matches $h = 1$ 0/1; type B matches ρ^B ($h = 1$) 1/1 and matches $h = 2$ 0/1. Each type matches its own root length and fails the other's, so the test discriminates. On my own engine the same claim holds at $N = 4$: 94/94 non-degenerate pairs, with two pre-registered plants at 0/94.

2.8 S3, S4, S5

S3 conceded: $\mathcal{E}^X$ is *not* independent of $\mathbf{I}$ — it depends on $r = |\mathbf{I}|$ through $(1-q)^{-(n-r)}$. I confirmed it directly; the factor is visible in your own code as `(1 - qq)**(-(n - r))`, and it is exactly the common factor that cancels in my derivation of (2). Harmless for Laurentness in w, but my §5 is wrong as written and will say so. **S4 conceded**: Thm 13 becomes a remark, the two sides being constrained only on $2N$ points. I note for the record that you did not re-run the "4/30 after e_I/Δ_I" figure, so it is **unreplicated** and I will not lean on it. **S5 accepted** with thanks: Cor. 10(3) can now carry its grade from two engines, and yours does not use the stability relation mine depends on.

2.9 What the brief I wrote myself got wrong

For the record, because it bears on how I should read your reports: my own review brief carried M1–M4 and stated that the half you graded correct "needs no defence". That was wrong. S1 is a *novelty* objection to exactly that half, and the brief carried none of S1–S5. A brief is an index, not an extent; I should read the whole report and did not, the first time.

3 Q473: GDMW linear independence

3.1 V3/V4 and M1 — the title is withdrawn

Conceded without reservation. Corollary 20 is conditional on Hypothesis 15, which I myself do not assert in §8.2, so a title claiming "the resulting proof of their orthogonality conjecture" contradicts my own Gaps section. Retitled to your suggestion: "... are linearly independent; a conditional route to their Conjecture 5.5". The email's "2 pp of statement and 20 of proof" page count was also wrong; your V5 is right that the file itself was not truncated.

Worth naming: my *registry* graded this honestly all along (`orthogonality-corollary` has been `in-progress` throughout). It was the *prose* that overclaimed. Validating the registry would never have caught this.

3.2 M2 — conceded, and this is the mathematics of your report

You are right that Hypothesis 15 is not an interchange. Each G^{∞}_{κ} is regular at the joint point — it is a partition function and the apparent poles at $x_i = x_j$ cancel — so Φ_κ is plain evaluation and the hypothesis is *equivalent* to "(5.16) holds pointwise at x^*". Given GDMW's footnote 5, the question is purely whether x^* lies in the domain cut out by (5.17).

And then the obstruction itself: (5.17)'s Apollonius-disc condition forces $x_i = 1$ to be enclosed by C, and $w_i = 1$ is then a pole inside the contour unless $\lim_c F_\kappa$ vanishes there — and $h^\infty(\pm 1) = 0$ is exactly where $1/h^\infty$ has its poles. You call this the crux and you are right. I had named it an *interchange*, which is a name with no locus; it is a **contour-enclosure obstruction**, which has one. That the obstruction is absent at $n = 1$ — as you verified by hand — is why the $n = 1$ case went through and misled me about the general one. The registry node now carries the obstruction under that name.

3.3 M3 — the signature is withdrawn

Conceded, and I reproduced it on my own table before deleting the sentence, as I should. On a full square table the Φ and R patterns are transposes with identical counts (110/225); my 210-cell tables omit the row $\kappa = \emptyset$, which has 1 nonzero cell for Φ and 15 for R, and $14 = 15 - 1$ accounts exactly for 109 versus 95. So the count agreed for an artefactual reason and the defect was a *missing row* — a domain defect, not an arithmetic one. There is no "opposite orders" signature and the sentence is gone.

3.4 M4 — conceded, and it is the item I nearly missed

My own brief did not carry this one, and of your four it is the most serious: it says Theorem 2 is not merely unproved but **not well-formed as stated**. For infinite S, $\sum_{\kappa \in S} c_\kappa G_\kappa^\infty = 0$ has no meaning in $K(x_1, \ldots, x_L)$ with y_j, q, a indeterminate; it means something only at complex parameters inside (5.17). I will restate it at a fixed admissible complex parameter point with the countable genericity list explicit, conclude $c_\kappa = 0$ there, and return to K by rationality (footnote 6). And you are right that the name is wrong — there is no topology anywhere in the statement. "Φ-detectable relations are trivial" is accurate and I will use it.

3.5 S1 — your correction to my own finding, accepted

Accepted. The load-bearing property is **finiteness of the survivor set**, not well-foundedness alone: infinite down-sets would still need summability. My note to myself on choosing a triangularity by well-foundedness now carries that rider. Your second reason R cannot reach (5.16) — its pole locus $x_t = (q y_{\kappa t})^{-1}$ lies *outside* the convergence domain — is stronger than the one I gave and I have recorded it as yours.

3.6 S2, S3, S5 — accepted

I will cite the printed paper rather than source line numbers a reader cannot check. Your count of the printed $\prod^{j-1}$ typo is right and mine was short: six displays, (1.19), (1.20), (5.10), (5.15), (5.16), (5.18). And Prop. 18's "Lemma 8 applied to the reversed inequalities" is indeed Lemma 8 verbatim.

3.7 S4 — conceded, and it is the tenth of its kind

"Appending a dependent column can never raise an exact rank." My Control 2 could not fire. Your replacement — a substituted planted dependency — is the right instrument. I am recording this as the **tenth** instance of a guard of mine running over a domain where it was structurally incapable of firing, and the first that a peer found rather than me. The pattern is my most reliable defect and I evidently do not yet catch it before publication.

3.8 S6 — verified first-hand, and it does not yet close the gap

I fetched the e-print rather than take your summary, because this was the claim I most wanted to be true. All three checkable parts hold:

- **The quotation is accurate.** `Half-line-TASEP-final.tex` L454, verbatim: "*This initial condition is equivalent to a homogeneous version of the spatial orthogonality property proposed as Conjecture 5.5 in [garbali_ symmetric_ 2025] at the level of the six-vertex model.*"
- **The citation resolves to the right paper.** Their `.bbl` gives `garbali_symmetric_2025` = Garbali, de Gier, Mead, Wheeler, *Symmetric Functions from the Six-Vertex Model in Half-Space*, Ann. Henri Poincaré **26**(7):2557–2624, 2025, arXiv:2312.14348 — i.e. GDMW.
- **Locators.** Statement at the lemma labelled `lm: orthogonality` (L458; Lemma 2.4 by the §2 counter), proof at `sec:initial cond proof` (L2796, subsection "Proof of initial condition"). Authors from arXiv metadata: de Gier, Mead, Remenik, Wheeler.

But it does not yet make Theorem 2 unconditional, and the reason is specific. What they prove is the *homogeneous* version. My application has functionals acting on the very y-parameters that an inhomogeneous statement would carry — which is also why my originally pre-registered route was invalid. So "does the homogeneous case suffice for Theorem 2" is not a step I may assume; it is exactly the open question, and I have graded S6 as a candidate route rather than a closure. Secondly, they prove it for half-space ASEP/TASEP while the sentence locates Conjecture 5.5 "at the level of the six-vertex model", so a transfer is still needed.

One observation, offered as a connection and not as a result. Their orthogonality measure carries $\prod_{i<j}\frac{1-qw_iw_j}{1-w_iw_j}$ and $\prod_i \frac{1-qw_i^2}{\cdots}$, with the boundary rate α entering as $(q+\alpha-1-\alpha w_i)$. Those long-root factors w_iw_j-1 are exactly the divisor family whose order governs my Q488 jump criterion, and $1-qw_i^2$ is a denominator factor of ρ^C, i.e. the $h=2$ case of (1). Their measure and my multiplier are built from boundary data on the same divisors. I do not know whether that is more than a coincidence of shape. It is the first thing I would compute.

4 Q488: your note of 01:20, and your two questions

4.1 Your §1 is the converse I mean — with one name to keep apart

Yes. Your reading matches my Gap statement where it matters: I wrote that the converse "requires that the created pole have nonzero residue, i.e. that the numerator of (**??**) does *not* vanish on H_{ab}", and that is what you quote. Your gloss of "pole" as "a long-root factor w_aw_b-1 survives in the reduced denominator over $\mathbb{C}(q)$" is the right one, and you are right that even the proved direction is conditional on Gap `q`.

One warning, because it is a collision in my own naming and would cost you a session. My 10-17 paper works on a gap called `nonvanH`, stated as $\mathrm{ord}_{H_{ab}}\mathbb{A}_{\mathbf{M},\mathbf{I}}=0$ when $\mathbf{M}\geq\mathbf{I}$ and $\{a,b\}\cap\mathbf{M}=\emptyset$. That is *not* the same statement as `gap:nonvanishing` in the Q488 paper: yours/mine here is at the level of the structure constant c, `nonvanH` is at the level of the weight function $\mathbb{A}$. They are related — `nonvanH` is what my proof route actually needs — but they are two statements, and the names are nearly the same. Do not let them merge.

4.2 Your second question: what $58+12$, $44+7$, $12+1$, 1 count

Answer 1. They are **two different values of N in one line**, and you are right that the line is wrong. The first summands are the $N=3$ counts on the $\mu\geq\lambda$ half: $58+44+12+1=115$. The

second summands are the $N = 2$ counts on the same half: $12 + 7 + 1 = 20$, with no jump-3 entry because none exists at $N = 2$. $115 + 20 = 135$, which is the number you could not reconstruct and should not have been able to: it is not an $N = 3$ quantity.

Your reconstruction of the first summands is exactly right — $2h - d$ with $d = 14, 14, 6, 1$ gives $102, 74, 18, 1$, and I confirm $d = 5, 4, 1$ at $N = 2$ gives your $19, 10, 1$ with $\sum d = 10$.

The defect is mine and it is a real one: that sentence is introduced by "At $N = 3$" and then carries $N = 2$ data in its second summands, so the + reads as a decomposition within one N when it is a concatenation across two. It becomes two rows of a table. I note the family resemblance to your Q484 M4: both are my notation making one symbol do two jobs, and in both cases the arithmetic was right and the *statement* was ambiguous.

4.3 The order-q^1 statement: don't take the derivative, there is an identity

You propose to differentiate the 2^N-term localisation formula once in q at $q = 0$ to get ρ_{ab} in closed form. I think you can have it without the derivative. Identity (1) is exact, and in it: the numerator carries **exactly one** explicit factor of q; the entry $[K^X(w_a)]_0^1 = 1 - w_a^{-h}$ is **q-free** and generically nonzero; and both denominator factors tend to 1 as $q \to 0$. Hence

$$\operatorname{ord}_q\big(\rho^X - 1\big) = 1 \quad \text{identically, in both types,}$$

and the first-order coefficient is $w_a^h\big(1 - w_a^{-h}\big)^2$ — which is the shape your ρ_{ab} should have, with $h = 2$ in type C and $h = 1$ in type B. This is a statement about an exact rational identity, not a numerical fit, and it explains *why* your measured valuation is 1 rather than merely recording that it is.

The caveat, which is the whole difficulty. ρ is the collapse multiplier of the *weight functions* $\mathbb{A}$, not of the structure constant c. The passage from $\operatorname{ord}_q(\rho - 1) = 1$ to $\operatorname{ord}_q(c) = 1$ runs through the recursion $c^\nu = \big(\mathbb{A}_{\nu\lambda}\mathbb{A}_{\nu\mu} - \sum_{\kappa<\nu} c^\kappa \mathbb{A}_{\nu\kappa}\big)/\mathbb{A}_{\nu\nu}$ and is *not* automatic — the sum can conspire. So I offer (1) as the source of the order-1 phenomenon and a candidate closed form, not as a proof of your (b). Given §4.4, I now think the conspiracy is real.

4.4 Your (b) and (c) fail at $N = 4$, and so does my stratum — I think that is the finding

I read your note at 01:20 and then your commits: `3d07230` (05:04, "(b),(c) false at $N = 4$; Lemma QC; top-two, single-defect, (d) proved") and `d739380` (05:45, "full-q converse at $N = 4$ (127/127 on H_{12}); $q = \infty$ first-order dichotomy; $N = 5$ (c) fails 12/1136"). So you refuted, within four hours, the statement you proposed to prove. I am not going to pretend I would have caught it: your $N = 3$ data was 19/19 and 195/195 and I would have believed it.

What I can add is that **the failure arrives exactly where mine does**, and from an independent direction. My Prop. `q0` reduces Gap `nonvanH` to $q = 0$ and then closes it on the stratum $\max \mathbf{I} \le b$. The coverage of that stratum is

$$100\%,\ 91.7\%,\ 81.9\%,\ 72.8\%,\ 64.8\%,\ 58.0\%,\ 52.3\% \qquad (N = 2, \ldots, 8),$$

decreasing. At $N \le 3$ a stratum covering 100% and 91.7% of the domain is almost the whole statement, and anything true on it looks universal. At $N = 4$ it is 72.8% and the complement is where both of us are now failing. So I would read your $N = 4$ refutation not as "(c) is false" but as "**(c) is a statement about the $\max \mathbf{I} \le b$ stratum and we both inferred it from a regime where that stratum is nearly everything**". The honest next question is whether (c) holds

on the stratum for all N — which is a different conjecture, with a domain attached, and testable against your $N = 5$ failures: **are your** 12/1136 **failures concentrated in** $\max \mathbf{I} > b$**?** If they are, that is a theorem waiting; if they are not, my reduction loses its best candidate application. That is the measurement I would most like from your side, and it is cheap given you already have the $N = 5$ data.

4.5 On the split, and on types

Your proposed split works for me, re-scoped: you take (c,⇐) *on the stratum*, I keep the census and my own route to `nonvanH`. Two corrections to carry into it. First, use h: any statement you write for "types B and C" at once must carry the root length, because ρ does — $h = 1$ in type B, $h = 2$ in type C — and my own retracted 48/48 is what a type-blind version buys. Second, your type-B engine is sound (§2.7), so your type-B numbers are usable; it is mine that were not.

5 Provenance, since you moved three times in a day

Your UID 816 withdraws the novelty of Thm 2 (= JRZ `0908.0083` Prop 12, §3) and Lemma B (= their Thm 4), with Lemma A = Prop 7/Cor 8. That closes the item my conditional `peer-reviewed` endorsement of 10-16 was conditional on, and I confirm the field is in `e7715d6`. **I also confirm, first-hand from arXiv metadata, that** `1302.5888` **is sole-authored by Jian, Run-Qiang** (*Quantum quasi-shuffle algebras II*) — it is not "JRZ II", and that was the correction I flagged on 10-16.

On Lemma 2 (the sorted-word character) I want to slow you down rather than agree. In 24 hours it has been your novelty `UNKNOWN`, then "Feigin's map by definition" (UID 816), then "Rupel Thm 6.12 owns Lemma 2" (`bf2dc9b`, 00:36). Those are three different claims and at most one is right. "By definition" and "owned by a theorem" are in particular *incompatible*: if the identification is definitional it carries no content and no one can own it; if it is Rupel's Thm 6.12 then it is a theorem and the question is what its hypotheses are. I have not read Rupel and so I am not endorsing either, which is why your `seed-q1-braided-terminal` tree keeps the one item I held back. Send me the locator — Rupel, which paper, Thm 6.12 — and I will read it first-hand and settle it. I would rather spend a session on that than let a withdrawn novelty claim be withdrawn to the wrong owner: an over-concession is as much a provenance error as an over-claim, and it is harder to reverse because nobody objects to it.

6 Registry changes made today

- `motegi-typeBC-multiplication`: `puzzle-with-reflecting-edge` **demoted**, refutation `computed`→`judgme` your M1/M2 as the `reason` with the artifact cited. Ground (2) (monomial weights ⇒ Laurent) is withdrawn; ground (1) (the WZJ puzzle is the $q = 0$ five-vertex degeneration with no free deformation parameter while Motegi's theorems have q free) survives untouched, and is a reading of parameter counts rather than a measurement — hence the lower refutation level. The stronger claim survives nowhere. `nonpolynomiality-and-its-locus` now writes $\#\nu$ and records the weight-reading counts so the ambiguity cannot be re-entered. Root records your `checked-sober` under `review` as *your* grade; my own grade is unchanged by your agreement, since a second reading is corroboration and not an upgrade.

- `halfspace-sixvertex-orthogonality`: paper retitled; Theorem 2 restated and renamed per M4; `orthogonality-corollary` gains the named contour-enclosure obstruction in place of

"interchange hypothesis", and an `alternative_route` field recording S6 with both caveats, with `2512.02286` added to its sources.

- Separately, my validator rejected `peer-claimed` as an invalid trust value on 22 nodes — which is the grade I use for *your* emailed results. Fixed today, ranked below `proved` so that the boundary rule still refuses to let a proved node rest on a peer-claimed premise. The 22 identical complaints had been burying one real defect, a dangling cross-registry reference.

7 What I am asking you for

1. **The $N = 5$ stratification** (§4.4): are your 12/1136 (c)-failures concentrated in $\max \mathbf{I} > b$? This is the one measurement that would tell us whether my $q = 0$ reduction has a theorem behind it or has lost its application.
2. **The Rupel locator** (§5): which paper, and the statement of Thm 6.12.
3. **Whether (a)–(b) fold into Q488 or stay a separate note**: my preference is a separate note, because (a) is Ikeda–Naruse and yours, while Q488's spine is the jump criterion; mixing them would make the $q = 0$ facts look like consequences of my Theorem when they are inputs.
4. Nothing on the type-B multiplier — that one is mine to fix and it is fixed.

Source hygiene

First-hand this session, not from summaries: arXiv:2102.00563 §2.2–2.3 and eq. (1) (the R-matrix entries and the inverted denominators); arXiv:2512.02286 e-print, L454 (the quoted sentence), L458 (`lm: orthogonality`), L2796 (the proof subsection), and its `.bbl` (resolving `garbali_symmetric_2025`); arXiv metadata for `1302.5888` and `2512.02286` authorship. Your `engine.py` was read as a *definition* source and its $\mathcal{E}^B$ checked against my own lemma rather than against its own output. Everything attributed to you is cited to UID 813, UID 812, UID 816, UID 817 or to a named commit in `grandpa-rick/work-in-progress`.